NeuroMaze: Designing an EEG-Based Interactive Maze Game

Abstract
This project develops a maze game controlled by EEG data, where the player's attention level influences the character's speed and environmental interactions. EEG attention values affect the player's movement and trigger dynamic effects in the environment, such as wall "breathing" and transparency changes in goal objects. By using real EEG data, the project offers a unique interactive experience, where concentration directly alters gameplay and the environment.

CCS Concepts

- Human-centered computing → Human-computer interaction (HCI)
- Applied computing → Interactive art
- Hardware → Brain-computer interfaces (BCI)

Keywords
EEG Data, Game Design, Immersive Experience, Interactive Design, Real-time Data Processing, Maze Game

Introduction
This project explores a maze game controlled by EEG data, where the player's attention determines movement speed and interacts with the environment. The game adjusts the character's speed and dynamic effects like wall "breathing" and transparency changes based on EEG data. This interaction enhances immersion by allowing players to control the game with concentration.

Objectives
The goal is to create an interactive maze game where EEG data controls both the character's behavior and the game environment, enhancing immersion through real-time interaction.

Background
EEG technology has been used in gaming to adjust game difficulty or character movement based on brainwave signals. This project extends this concept by dynamically altering the game environment, offering a more immersive experience.

Techniques Used
The project uses the Ursina game engine, Python, and NumPy for data processing and integrating EEG data to control the game's dynamics.

Technical Implementation

- **Data Processing**: EEG data is collected from CSV files. Noise and errors in the data are reduced using smoothing algorithms like moving averages.

```python
def load_eeg_data(file_path):
    eeg_data = []
    with open(file_path, newline='') as csvfile:
        reader = csv.reader(csvfile)
        header = next(reader)
        alpha_index = header.index('Alpha_TP10')
        for row in reader:
            try:
                eeg_data.append(float(row[alpha_index]))
            except ValueError:
                continue
    return np.array(eeg_data)
```

- **System Architecture**: The game is structured using OOP. Key components include:
    - Maze: Generates and manages maze structure and wall dynamics.

```python
def generate_maze(size):
    maze = np.random.choice([0, 1], size=(size, size), p=[0.7, 0.3])
    maze[0, 0] = 0
    maze[size - 1, size - 1] = 0
    return maze
```

    - Player: Controls player movement speed based on EEG data.

```python
player = Entity(
    model='sphere',
    color=color.blue,
    scale=1,
    position=(0, 1, 0),
    collider='sphere'
)
```

    - EEGData: Reads and processes EEG data for the game.

```python
for wall in walls:
    breath_value = np.clip(attention * 3, 0.5, 4)
    target_scale_y = tile_size * breath_value
    if np.isnan(target_scale_y) or np.isinf(target_scale_y):
        target_scale_y = tile_size * 1.5
    wall.scale_y = lerp(wall.scale_y, target_scale_y, 0.1)
```

    - Game: Manages game flow and player-environment interaction.

- **Agent Design**: The Player Agent adjusts speed and interacts with the environment based on EEG data. Higher attention speeds up the player, and the environment reacts with effects like "breathing" walls.

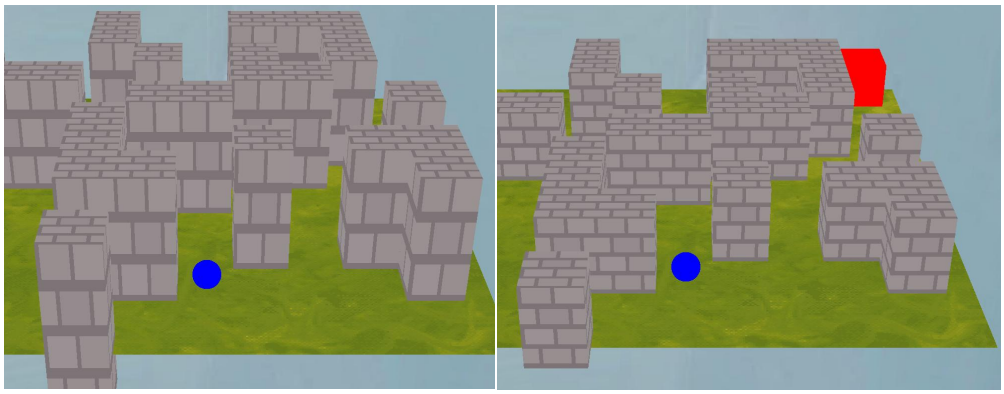
Process

The game design focuses on using EEG data for interaction. Attention levels control

the player's speed and alter the game environment. A smooth visual transition is achieved through linear interpolation for wall height changes, enhancing immersion.

Results

Testing showed that the player's speed and the environment's dynamic effects could be controlled via EEG data. Noise and data fluctuations were mitigated through smoothing algorithms. Feedback showed enhanced immersion as concentration levels changed the game experience.



Reflection

The project achieved its goal of enhancing interactivity using EEG data. Limitations include real-time EEG data acquisition and environmental changes that are relatively simple. Future improvements could involve additional EEG signals and more complex environment interactions.

Conclusion

This project deepened my understanding of integrating EEG data with game interaction systems. It showcased the potential of Brain-Computer Interfaces (BCI) in enhancing user experience through innovative technology.

References

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